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# The relationship between e-commerce industry cluster environment, entrepreneurial ability and entrepreneurial performance

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**Abstract**
With the continuous maturity of Internet technology, it has created favorable conditions for the rise of e-commerce industry. The lower entry barrier has significantly promoted the development of the industry. When enterprises gather to form industrial clusters, they can achieve better economic results due to spatial agglomeration. The clustering effect is the fundamental characteristic that distinguishes industrial clusters from ordinary regions. Enterprises can easily obtain sufficient resources, rich entrepreneurial information, and entrepreneurial environment that is more conducive to innovation, competition and development in the cluster area, which are concrete manifestations of the clustering effect of industrial clusters. Therefore, the purpose of this paper is to explore the development level of e-commerce industry on the basis of industrial cluster theory, to analyze the current situation of e-commerce development, and analyze the problems and solutions in the process of entrepreneurial ability and output value. This paper adopts a mixed-methods approach, combining semi-structured interviews and questionnaire surveys, to investigate the relationship between the e-commerce industrial cluster environment, entrepreneurial ability, and entrepreneurial performance. Hierarchical regression analysis is used to test the proposed hypotheses. The research results show that a mature e-commerce industry cluster system can reduce transaction costs, increase scale, and obtain cluster economies of scale and promote employment through effective integration of environment and goals. Furthermore, the environmental dimensions of e-commerce industry cluster have a significant positive impact on entrepreneurial performance and entrepreneurial ability, with the latter playing a mediating role. Therefore, fostering the sustainable and healthy development of the e-commerce industry, reduce unnecessary waste of resources, reduce the risk of personal entrepreneurship, enhance the scientific and technological strength of the industry, continuously sum up experience, actively improve and refine relevant methods, and jointly promote the rapid development of the e-commerce industry.

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**Keywords** E-commerce · Industrial cluster · Entrepreneurial ability · Entrepreneurial performance

## 1 Introduction

In the new economic era, the Internet economy, represented by "Internet +" and big data, is becoming an important part of the economic operation [1–3]. In recent years, with the rapid development of mobile Internet, e-commerce industry has achieved unprecedented rapid growth in entrepreneurial projects [4]. Driven by continuous technological advancement, the deepening development of global economy, the rapid increase in Internet users, and the diversification of demand, e-commerce enterprises are booming. However, widespread unemployment following the financial crisis has also motivated many to venture into e-commerce entrepreneurship. More and more people are employed in e-commerce enterprises, and various factors have contributed to the current e-commerce entrepreneurship boom. As a new business model, e-commerce not only offers cost-effectiveness and high efficiency, but also embeds start-ups into various business ecosystems, either actively or passively, gradually evolving into a more mature industry [5–6]. Therefore, to inject more vitality into the industry, it is necessary to investigate and study the information closely related to the entry threshold and performance of entrepreneurship [7].

With the application of mobile Internet, big data, artificial intelligence and other technologies, people's life style has changed dramatically. The emergence of e-commerce has changed people's consumption mode. Consumers no longer hand in money face-to-face, hand in delivery. The whole transaction can be completed through the Internet [8–10]. The number of registered e-commerce enterprises is increasing year by year. The rapid development of e-commerce enterprises is that they have low production cost, transaction cost and management cost, which improves the profits of the enterprises on the original basis. In recent years, there has been a wave of entrepreneurship in China. With the development of e-commerce, it provides people with new entrepreneurial options. At the same time, e-commerce entrepreneurship has the advantages of low entry threshold, low operating costs during the operation period, time-saving and labor-saving in daily operation, and less pressure on entrepreneurs, which makes most entrepreneurs keen on e-commerce entrepreneurship. At present, the overall situation of China's e-commerce industry is not mature. The main body of e-commerce entrepreneurship is still small-scale, small and medium-sized enterprises with generally insufficient business experience, and the number of successful enterprises is also very limited [11–13]. A key reason is that some enterprises fail to fully utilize surrounding resources through the cluster effect. Since Marshall introduced the concept of the industrial district in the 19th century, research on industrial clusters has continued. Simultaneously, extensive practice has demonstrated that industrial clusters serve as ideal incubators, providing entrepreneurs with essential resources and a favorable entrepreneurial environment [14].

A large body of research shows that the industrial cluster can effectively promote the development of entrepreneurial activities, and the cluster effect can bring entrepreneurial resources for enterprises in the cluster and create a high-quality entrepre-

neurial environment [15, 16]. Throughout the existing research, experts and scholars have conducted in-depth research on the impact of the environment and network in the industrial cluster on the performance of start-ups in the cluster [17, 18]. However, as the most prominent feature of industrial clusters, few scholars have explored the entrepreneurial performance from the perspective of industrial agglomeration effect. In addition, there is no discussion about the rural e-commerce industry cluster as the research object. Therefore, it remains to be explored whether industrial clusters can promote entrepreneurial performance and apply to e-commerce industry. The motivation of cluster is scale economy and external effect. Many interrelated enterprises gather together in geography, which makes the transaction cost between enterprises decrease, and reduces the production cost of enterprises. At the same time, the complementary and competition among similar enterprises make the enterprises in the cluster area form a joint force, produce greater competitive advantage, and obtain more industrial development. The research results of industrial clusters at home and abroad are also emerging [19]. Generally speaking, entrepreneurial performance is the main body of entrepreneurial activities, that is, the organizational performance of start-ups. The evaluation results of entrepreneurial performance will directly affect the evaluation and selection of a start-up's strategy and direction. Therefore, how to correctly measure entrepreneurial performance is an important topic of entrepreneurship related research. At present, domestic scholars study the dimensions of entrepreneurial ability from the perspective of entrepreneurs and entrepreneurial enterprises. However, there is no unified standard for academic division. Scholars are rooted in According to their own research purposes, we choose to use those dimensions to measure entrepreneurial ability. According to the existing division of industrial cluster environment, and combined with the characteristics of e-commerce industry, this paper puts forward the division of e-commerce industrial cluster environment dimension. Compared with the industrial cluster environment dimension, the main difference is to increase the logistics and other e-commerce industrial cluster environment, refine the infrastructure into network infrastructure, and at the same time, integrate the e-commerce industry Group environment, entrepreneurial ability and entrepreneurial performance are included in the same research model, which complements the research on the influencing factors of entrepreneurial performance in e-commerce industry cluster. Therefore, a correct analysis of the current development prospects of e-commerce industry in China is conducive to combining the advantages of industrial clusters, analyzing entrepreneurial ability and related theories, clarifying the entrepreneurial performance evaluation standards, and promoting the sustainable development of e-commerce industry in China [20–21].

Starting from the meaning and characteristics of e-commerce, this paper explores the development process of e-commerce industry, expounds the industrial cluster environment, entrepreneurial ability and performance of e-commerce enterprises, mainly analyzes the existing problems of e-commerce industry entrepreneurship, finds out reasonable solutions and the balance basis point in line with the Internet economy, and organically combines the two. On the basis of combing industrial cluster effect, Network Entrepreneurship and other related theories, this paper attempts to explore the problems existing in e-commerce industry and the specific application of Countermeasures in the process of entrepreneurship from the application level,

and comprehensively summarizes the reasons for low efficiency, low cost-benefit ratio, etc., hoping to provide theoretical basis and innovative business model for the problems encountered by e-commerce operators and entrepreneurs, Clear solution direction. Based on the positioning of the current business mode and learning from foreign experience, this paper analyzes the similarities and differences in the development of e-commerce enterprises both at home and abroad through comparative advantage, studies advanced experience, puts forward improvement methods and paths, combines with the development of new ways, and finally puts forward a new model for the development of China's e-commerce enterprises, and puts forward some suggestions for promoting the vitality of China's new economic development. To address the research questions, we employ a problem-oriented analytical framework that integrates qualitative insights from interviews with quantitative data from surveys, utilizing hierarchical regression analysis for hypothesis testing.

## 2 Method

### 2.1 Core concepts

#### 2.1.1 E-commerce

E-commerce refers to the use of the Internet for all trade activities, that is, the Internet will be information flow, capital flow, business flow and part of the logistics to complete the connection of the new business model. Operating in the open environment of the Internet and based on server/client architecture, e-commerce utilizes computer technology, network technology, and remote communication technology to achieve electronization, digitization, and networking throughout the entire business process. In the context of big data, e-commerce enables consumers to participate directly in economic activities online and offline, moving beyond traditional store-based marketing. Using online trading platforms reduces operating costs for enterprises, expands market reach, improves transaction success rates, and saves time for customers, creating a win-win situation.The development of e-commerce is remarkably rapid and extensive, allowing consumers to search for virtually any product or service online. It covers a wide range of goods, from daily necessities like food, clothing, books, and household items to high-end products like consumer electronics and cars. Through e-commerce, consumers can shop from home via the Internet, effortlessly acquiring desired products.

#### 2.1.2 Industrial cluster environment

Industrial cluster was first proposed at the end of last world. A whole industrial chain includes production enterprises, suppliers, financial institutions, scientific research institutions and business support enterprises at all stages, forming a group in a specific region by virtue of the advantages of aggregation. In this environment, enterprises can reduce R & D costs, set reasonable goals, improve efficiency and reduce waste, integrate technology, funds, policies and talents, so as to achieve economies of

scale. Industrial clusters have become the main driving force for the development of regional innovation economy at home and abroad. E-commerce industry has a high degree of information gathering and a strong scale effect. By effectively organizing the internal operation process of enterprises and adjusting the mutual functions with the help of the industrial cluster environment, it can continuously focus on the development of advantages. According to the theory of industrial cluster development, e-commerce industrial cluster can be divided into three levels. The first layer belongs to the industry core layer, including the core industry chain and other related peripheral activities, such as payment and settlement industry, supply chain industry, software service industry, media exhibition industry and other upstream and downstream activities. The second layer is the cluster support layer, which mainly consists of logistics funds, financial management and other support organizations; the third layer is called environmental policy layer, which mainly consists of government related industrial policies, talents, science and technology and other related organizations. Improving the environment of e-commerce industry cluster can improve the cross organization sharing and information gathering of e-commerce, reduce the cost of large-scale production and improve the efficiency of e-commerce activities, which is an indispensable step for the development of the industry.

### 2.1.3 Entrepreneurship performance

Entrepreneurial performance is one of the important factors to measure entrepreneurial achievements, which is not only the concern of managers, but also the important understanding object of entrepreneurs. Generally speaking, entrepreneurial performance is the main body of entrepreneurial activities, that is, the organizational performance of start-ups. The evaluation results of entrepreneurial performance will directly affect the evaluation and selection of a start-up's strategy and direction, so how to correctly measure entrepreneurial performance is an important topic of entrepreneurship related research. At present, the research on entrepreneurial performance is mainly based on the theory of organizational performance. In addition, there are still differences in the definition of entrepreneurial performance, which can reflect the different growth indicators of an organization by using different indicators to measure entrepreneurial performance. In order to fully reflect the enterprise performance, profitability index can also be added to reflect the future profitability of the enterprise. At present, whether the start-ups in the e-commerce industry cluster can survive and gain competitive advantage in the early stage of entrepreneurship and maintain long-term healthy development is the key problem faced by the start-ups. Therefore, this paper combines profit indicators and growth indicators to measure performance, and uses the growth performance and profit performance measurement methods of new enterprises for reference.

### 2.1.4 Entrepreneurial ability

Entrepreneurship is a process from the start-up period to the growth period. The growth period is usually affected by internal and external factors of the enterprise, which is more challenging than the start-up period. Therefore, during growth, enter-

prises must focus on external factors and the enhancement of their own capabilities to sustain entrepreneurial momentum.In this study, Entrepreneurial Ability refers to the capacity of entrepreneurs or entrepreneurial teams to identify and exploit opportunities, integrate resources, and achieve sustainable performance in the process of venture creation and growth. It is conceptualized and measured along three dimensions. The first is opportunity identification ability which is the capacity to perceive and discover potential market opportunities. The second is resource integration ability which is the capability to mobilize and combine human, financial, and technological resources. The third is relationship management ability which is the skill to build and maintain relationships with stakeholders such as customers, suppliers, and government agencies. These dimensions are measured using a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree) based on self-assessment by entrepreneurs and managers in the surveyed e-commerce start-ups. The measurement scale was developed based on literature review and preliminary interviews, demonstrating good reliability and validity.

## 2.2 Enterprise e-commerce entrepreneurship mode

### 2.2.1 B2B mode

B2B(Business-to-Business) mode involves transactions between enterprises facilitated by Internet technology. Through online trading platforms and collaborative work platforms, it provides transaction intermediation, communication channels, and business operation tools for both parties to conduct business cooperation. After more than a decade of development, major e-commerce enterprises in China have evolved into comprehensive platform. These are not merely transaction venues but integrated e-commerce enterprises with complete facilities, extending the entire transaction process forward and backward. They integrate enterprises involved in transactions, consumers, logistics companies, banks, government departments, and customs, creating a virtual comprehensive enterprise centered on resource sharing and connected by information, thereby enhancing the core competitiveness of both the enterprise and the e-commerce industry.

### 2.2.2 B2C mode

B2C (Business-to-Consumer) mode involves enterprises or businesses directly selling to consumers, recommending products, business activities, financial projects, services, and other comprehensive activities to individuals through online information. This is the most common e-commerce model, typically manifested as online retailing of goods such as books, flowers, computers, and communication supplies. Currently, domestic e-commerce enterprises provide varied service content based on their positioning. Enterprises using this model can be classified into E-brokerage, online direct sales, e-retail, distance education, online entertainment, online booking, online finance, and other types, comprehensively meeting consumer needs.

### 2.2.3 B2G mode

B2G (Business-to-Government) mode covers various interactions between enterprises and government, such as procurement, taxation, commodity inspection, and policy release. This mode encompasses a wide range of activities beyond mere transactions, including but not limited to government procurement, tax filing and payment, customs declarations, commodity inspection applications, license renewals, and the release of policy and regulatory information. A key feature of the B2G mode is its reliance on dedicated and highly secure e-government platforms established by public authorities. Through these platforms, governments publish tenders and requests for proposals, enabling businesses to access procurement opportunities, submit bids, and complete transactions electronically. This digitalization streamlines traditionally paper-intensive processes, significantly enhancing transparency, reducing administrative burdens, and promoting fair competition.

### 2.2.4 C2B mode

C2B (Consumer-to-Business) mode is a newer e-commerce model, different from B2C. It first emerged in the United States and has grown into two mature business models through rapid development. One model involves aggregating users to form a powerful buying group, thereby changing the weak negotiating position of individual users in one-to-one bidding. This allows consumers to purchase single items at wholesale prices, saving costs and gaining benefits. Taobao, for instance, provides such a group-buying platform. The other model leverages the aggregation of many users to adjust production according to market demand, offering personalized customization services tailored to consumer preferences. In this model, consumers contribute ideas, and enterprises turn them into reality.

## 2.3 Special characteristics of e-commerce industry entrepreneurship

### 2.3.1 Ecology

The speed of information and resource circulation among e-commerce enterprises is faster than that of traditional enterprises, and the competition is also extremely fierce. Every enterprise is trying to adapt to the surrounding environment, so as to build a constantly improving, high-speed development of e-commerce ecosystem based on the Internet. This system is characterized by openness, complexity, richness and coevolution. The ecological nature of e-commerce entrepreneurship is to cooperate with each other and complement each other in business activities. The system has formed increasingly close and diversified ecological relations, which is indispensable. E-commerce enterprises can share production and sales experience with cooperative companies, learn from each other's strengths and weaknesses, and even establish business alliances, and often participate in business alliance activities to outsource network marketing work to other e-commerce enterprises to cooperate in the field of fund borrowing.

### 2.3.2 Personalization

As e-commerce provides a wealth of communication tools, such as forum, Alibaba Wangwang, huicongfa, blog class, in station letters, etc., which further shortens the distance between the buyer and the seller, e-commerce start-ups can collect demand information in time and provide more personalized products and services. E-commerce entrepreneurship draws on the characteristics of e-commerce itself to make a personalized entrepreneurship plan. Consumers' love for e-commerce is due to the strong service and pertinence, which is easy to be suitable for businesses. Entrepreneurs choose work place, work method and work personnel according to their actual situation, which is less restricted by the environment. Through point-to-point online communication and low-cost special design, we provide consumers with personalized products and services.

## 3 Experiment

### 3.1 Data source and research design

This study employed a sequential mixed-methods design. First, a qualitative phase involving semi-structured interviews was conducted to refine the research model and measurement items. This was followed by a quantitative phase using a survey questionnaire to test the hypotheses. We selected e-commerce start-ups from four major cities in China (Beijing, Hangzhou, Shanghai, Guangzhou), which are recognized as the most active e-commerce hubs. A stratified random sampling method was used to ensure diversity in industry sectors, development stages (creation, growth, expansion), and profit models. The interviewees and survey respondents involved high-level management, middle-level management and grass-roots employees. In total, 200 questionnaires were issued, and 180 effective questionnaires were recovered, yielding a valid response rate of 90%. The content includes the basic situation of its operation, the environmental situation of the industrial cluster, the entrepreneurial ability and the relevant performance evaluation standards. All the experimental data sources of this study are obtained. The questionnaire used a 5-point Likert scale and was administered online. Data analysis was performed using SPSS 26.0, involving reliability analysis, validity analysis (including factor analysis), and hierarchical regression analysis to test the mediating effects. (Table 1).

### 3.2 Research process

First of all, through the semi-structured interview method, the entrepreneurs and their employees of typical e-commerce start-ups in different development periods, different industries and different profit models are interviewed and summarized. It is found that e-commerce entrepreneurial activities or tasks have three main characteristics: virtual operation, technical support and distributed collaboration. E-commerce entrepreneurial pressure mainly comes from four aspects: individual endogenous, individual exogenous, organization endogenous and organization exogenous, and among

**Table 1** The regression analysis of cluster environment on entrepreneurial ability or performance

| | | Dependent Variable | | Relational Competence |
|---|---|---|---|---|
| | Variable | Model 1 | Model 2 | Model 3 |
| Control Variable | Education Level | −0.096 | −0.022 | −0.040 |
| | Entrepreneurial Experience | 0.111 | 0.092 | 0.061 |
| | Overall Environment in the Cluster | 0.265 | 0.189 | 0.277 |
| Independent variable | Education and Training | 0.213 | 0.309 | 0.309 |
| | Financial Support | 0.243 | 0.196 | 0.196 |
| | Logistics Service | −0.092 | −0.085 | −0.120 |
| | F Value | 7.928 | 3.651 | 16.913 |

different individuals or organizations, the focus of these pressures is likely to be quite different.

In the second study, the definition and validity of the theoretical concept model of e-commerce entrepreneurial characteristics and the structure of entrepreneurial stressors are verified. The research designed a special questionnaire based on the relevant literature and interview results. In the e-commerce start-ups that meet the definition of this study, we conducted two sampling surveys, and made statistical analysis of the final effective questionnaires.

The third study explores the evolution of e-commerce entrepreneurial pressure in different stages of entrepreneurial development. The research adopts the research method of on-site quasi experimental design, and uses the e-commerce entrepreneurship stress scale verified by research II, as well as the previous related personality, control source and organizational support scales. In the current most active e-commerce entrepreneurship practice in China's network service, information service and professional service industries, each randomly selected is in the creation stage, growth stage and expansion stage respectively Zhang's three e-commerce start-ups were investigated and 180 valid questionnaires were statistically analyzed.

### 3.3 Purpose of the experiment

Through the interview method, the interviewers conduct a comprehensive and in-depth observation on the interviewees through the oral communication with the interviewees, and learn the motivation, attitude, personality and values of the interviewees, so as to collect the relevant first-hand information. Based on the limitations of the current literature research on the training mechanism of cross-border e-commerce talents' Entrepreneurship competency, this interview study explores the characteristics of cross-border e-commerce talents' Entrepreneurship competency and the composition and operation mechanism of the training mechanism from the empirical level. Mainly through interviews with cross-border e-commerce talents and relevant experts involved in talent training, this paper analyzes the growth experience of cross-border e-commerce talents, and the training elements that affect the development of entrepreneurial talents' competency, so as to obtain relevant data, refine the

components and characteristics of cross-border e-commerce talents' entrepreneurial competency through analysis, and the composition of cross-border e-commerce talents' entrepreneurial competency training system, as well as the influence of various training elements on the entrepreneurial competence of cross-border e-commerce talents. In order to try to build a preliminary theoretical model of the training mechanism of entrepreneurship competency of cross-border e-commerce talents. The results show that the environmental dimensions of e-commerce industry cluster have a significant positive impact on the entrepreneurial performance of enterprises. The environmental dimensions of e-commerce industry cluster have a positive impact on the dimensions of entrepreneurial ability. The dimensions of entrepreneurial ability have a significant positive impact on entrepreneurial performance. The dimensions of entrepreneurial ability exist in the relationship between the environment of e-commerce industry cluster and entrepreneurial performance In intermediary work use.

## 4 Discussion

### 4.1 Detailed discussion of results

The regression results provide nuanced insights into the relationships between cluster environment, entrepreneurial ability, and performance. The overall cluster environment, education and training, and financial support show significant positive effects on both entrepreneurial ability and performance (Model 1 & 2). This underscores the critical role of a supportive ecosystem and 'soft infrastructure' in nurturing entrepreneurial capabilities and outcomes.

However, the effect of logistics service on relational ability and opportunity ability is not significant and even slightly negative. A possible explanation is that in mature e-commerce clusters, logistics services have become highly standardized and commoditized. While essential, they may no longer serve as a differentiating factor that directly enhances an entrepreneur's specific abilities like relationship building or opportunity identification. Its impact might be more indirect, forming a baseline necessity rather than a direct driver of advanced entrepreneurial competencies.

Furthermore, the findings confirm the mediating role of entrepreneurial ability. The cluster environment enhances entrepreneurial performance not only directly but also indirectly by fostering the development of entrepreneurs' opportunity identification, resource integration, and relationship management skills. This highlights that investing in the cluster environment ultimately pays off by cultivating more capable entrepreneurs, who in turn achieve better performance.

### 4.2 Problems in e-commerce industry

#### 4.2.1 Network customer management pressure

The combination of Internet and commerce greatly facilitates the choice of consumers, and at the same time, it also stimulates more demands of consumers. Consumers pursue better quality and cheaper goods and services. This is not only an opportunity

but also a challenge for e-commerce start-ups. Under the new situation, e-commerce enterprises are always thinking for consumers and meeting the interests of customers has become the key to survival and development. Therefore, the network customer management of e-commerce start-ups will face higher requirements and challenges. Because many consumers do not fully trust the quality and price of products or services provided by e-commerce, the sensitivity of most people to e-commerce is still relatively weak at present. Therefore, many enterprises have designed special software systems to serve and record the content and process of customers under the role of website customization technology services, so as to better grasp their personalized needs and improve their satisfaction and loyalty. Therefore, creating value for them is the key to network customer management.

### 4.2.2 Pressure of innovation and competition

The emergence of e-commerce industry breaks the limitation of time and space, makes the market more transparent and competitive, and makes information transmission more efficient and convenient through the network. As a new entrant of the market, e-commerce start-ups are difficult to copy the successful path of traditional industries. The past era of opportunities obtained through information inequality and regional restrictions is gone forever. In the new era, only by providing creative products or featured products and services can we win a place in the market. As a mature enterprise in the market for many years, many internal processes, business management modes, such as profit model, website style, order process, etc. are completely open on the Internet, which are easy to be imitated and utilized by the latecomers. Therefore, they have to maintain their leading position in the market through continuous service and technological innovation. As shown in Fig. 1, in recent years, the transaction scale of China's e-commerce market has been expanding, and the number of enterprises has been increasing, so the competition among enterprises is very

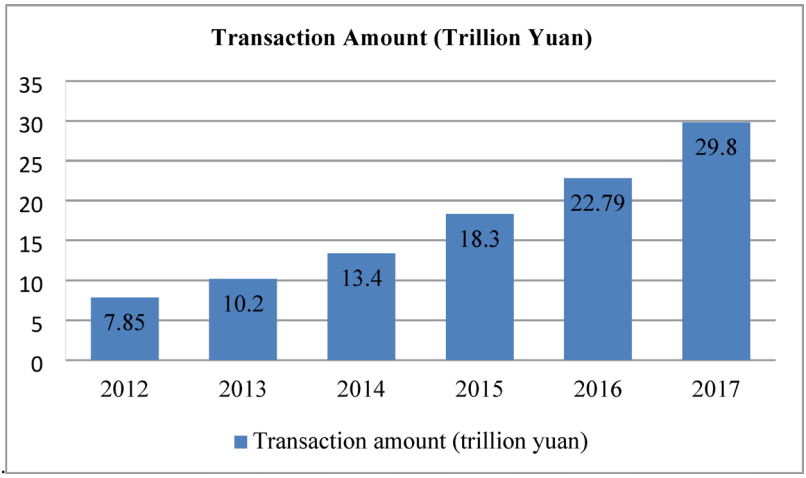



**Fig. 1** Change of transaction scale in China's e-commerce market from 2012 to 2017

fierce, and the innovation pressure has become a problem that every entrepreneur must face.

#### 4.2.3 Network trust pressure

In the virtual network environment, trust is no longer the primary premise of business activities. In the process of entrepreneurship and business operation, as long as there is a trust crisis, enterprises will not survive. At present, e-commerce enterprises also attach great importance to building a trust atmosphere, through the establishment of online transaction credit rating system, with the help of guarantee, the reputation of cooperative enterprises and after payment mechanism. Although many problems have been solved, the trust relationship with consumers has not been dealt with well, and customers are also very cautious about network security and data backup. At present, the development of e-commerce is slow, one of the main reasons is the lack of trust in individual transactions. As shown in Fig. 2, with the growth of the establishment time of the enterprise, different types of pressure change differently. Therefore, Taobao launched the "Alipay" service in the form of third party guarantees and gained the favor of consumers, making the volume of transactions increase at a geometric speed.

### 4.3 Suggestions and countermeasures for optimizing e-commerce industry

#### 4.3.1 Create a good environment for e-commerce industry

The research shows that the network infrastructure, education, training and financial support in the e-commerce industry cluster environment can positively affect the entrepreneurial performance, and can effectively promote the entrepreneurial activities of e-commerce related enterprises, so as to improve the entrepreneurial performance. Therefore, in the follow-up development of the cluster, we should focus on optimizing the internal environment of the cluster. It is necessary for the government

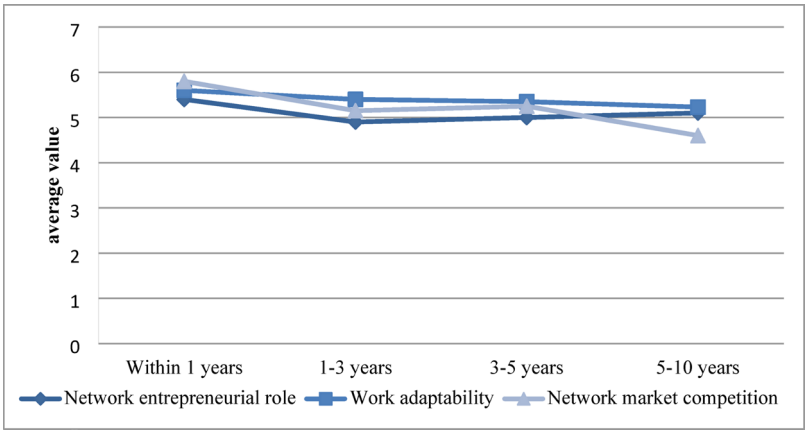



**Fig. 2** Pressure of e-commerce Entrepreneurship

and other relevant organizations to pay attention to the construction of e-commerce Industrial Park, provide financial support for the development of e-commerce Industrial Park, and make the optimization of the environment in the cluster free of worries. At the same time, the e-commerce industrial park should continuously upgrade the special line network in the cluster to provide guarantee for the enterprises in the cluster to maintain high operation efficiency. Regularly organize training activities for the enterprises in the cluster, provide a loose financing environment for the enterprises in the cluster, and provide material guarantee for the successful entrepreneurship activities of the enterprises in the cluster. At the same time, it is necessary to optimize the logistics service in the cluster, reduce the storage cost of enterprises, and provide more efficient distribution services for enterprises.

### 4.3.2 Emphasizing system reform

The innovation of the practical system of e-commerce industry for the cultivation of entrepreneurial ability is actually a process of system innovation and teaching innovation. The system reform can provide support for the innovation of teaching. It emphasizes that the system reform can promote the practice teaching of e-commerce to the greatest extent. It can also change the current situation of school management confusion under the mode of school enterprise alliance to the maximum extent. In terms of system, we should introduce preferential policies for entrepreneurs in the e-commerce industry. As shown in Fig. 3, the implementation of various policies can really improve the market survival rate. With the reform and development of the times, the innovation of e-commerce practice teaching for the cultivation of innovation and entrepreneurship ability has become an important measure in the process of implementing the curriculum core of relevant professional colleges. First of all, we should provide a solid guarantee in theoretical learning to reduce the obstacles in the process of entrepreneurship.

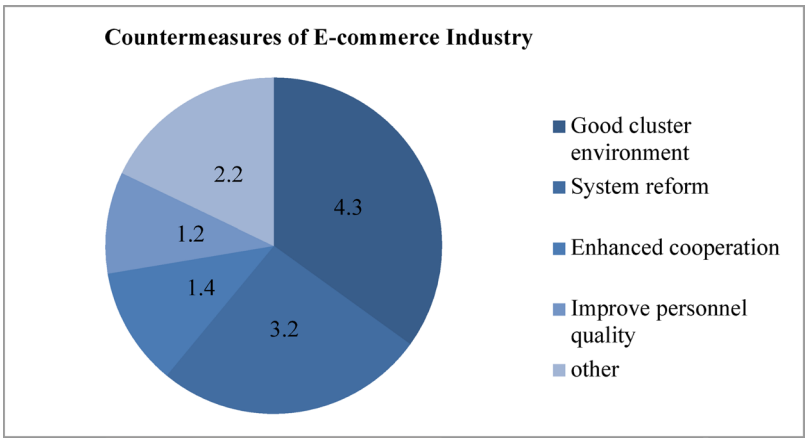



**Fig. 3** Countermeasures to optimize e-commerce industry

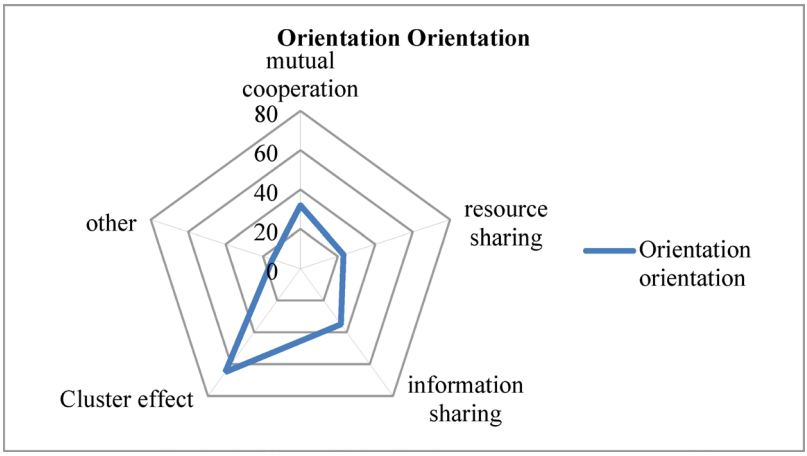



**Fig. 4** Orientation of e-commerce industry cooperation development

### 4.3.3 Guide the cooperative development of enterprises in e-commerce industry cluster

Cultivate a good cluster environment, gather the cooperation of e-commerce enterprises in the environment, give full play to the internal advantages of the cluster, achieve cost reduction, public infrastructure, benign competition, and ultimately maximize the benefits of consolidation. The heterogeneity of enterprise products in e-commerce industry cluster can not only show their own characteristics to maintain market diversification, but also complement each other's strengths and weaknesses. Combined with diversified resources, it can promote the differences between enterprises' benefits and ensure the sustainable upgrading and development of enterprise heterogeneity. Mature enterprises can play a leading role in the cluster environment and give full play to this advantage. As shown in Fig. 4, it is not only necessary to enhance the creation of cooperation and innovation oriented cluster culture, but also in other aspects such as resource sharing, information connectivity and other factors to enable enterprises in the cluster to learn and innovate technology together. Lead the new direction of future development.

## 5 Conclusion

In this paper, the research model of e-commerce industry cluster environment, entrepreneurial ability and entrepreneurial performance is mainly from the following four aspects: first, to test whether the impact of e-commerce industry cluster environment on entrepreneurial performance is significant. Second, whether the dimensions of e-commerce industry cluster environment positively affect the dimensions of entrepreneurship. The third is to verify whether the dimensions of entrepreneurial ability have significant impact on entrepreneurial performance. The fourth is to verify the intermediary effect of each dimension of entrepreneurial ability in the relationship

between the environment of e-commerce industry cluster and entrepreneurial performance. In this paper, the hierarchical regression method is used to analyze the research model of e-commerce industry cluster environment, entrepreneurial ability and entrepreneurial performance.

The cluster environment formed by e-commerce industry cluster can provide more complementary resources and a better and more relaxed entrepreneurial environment for entrepreneurs in the cluster. Resources and development environment are the most critical external factors that affect entrepreneurship performance, and to a large extent determine whether start-ups can develop healthily in the long term. A good internal and external environment is the guarantee to cultivate the entrepreneurial ability of enterprises. For the start-ups who are short of resources and lack of funds, a series of services and preferential measures provided in the e-commerce industry cluster can help the start-ups to a large extent, and gradually gain competitive advantages. At the same time, the empirical results show that the impact of logistics services on relationship ability and opportunity ability is not significant. The relationship and opportunity ability of start-ups can directly affect the performance of start-ups. New start-ups maintain good relationships with consumers, peers and relevant government departments, which helps enterprises to obtain resources more easily and identify potential opportunities.

This study has several limitations that offer avenues for future research. First, the sample is limited to four major, highly developed cities in China, which may affect the generalizability of the findings to rural or less-developed regions where cluster dynamics might differ. Second, the cross-sectional nature of the data prevents the establishment of causal relationships. Future research could adopt longitudinal designs to explore the dynamic evolution of entrepreneurial ability within clusters over time. Third, the use of self-reported data for both independent and dependent variables may introduce common method bias. Future studies could benefit from multi-source data collection, such as combining managerial perceptions with objective firm performance data.

With the vigorous promotion of the combination of science and technology with the Internet and the new direction of modern economic development, and the promotion of employment level in China, the development speed of e-commerce industry will be faster than ever. It is necessary for us to grasp the development context, study and judge the development trend of industry demand, take the initiative to seize opportunities, actively respond to challenges, and work hard for the modern e-commerce industry in China.

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